Yichi (Dios) Yang

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Education

New York University New York, US
Bachelor of Arts in Mathematics and Economics Theory, Minor in Computer Science. Sept 2019 – May 2023
- Overall GPA:3.50/4.00, Major GPA:3.55/4.00
- Main Courses: Ordinary Differential Equation, Real Analysis, Math Modeling, Math of Finance, Statistics and Probability, Econometrics, Time Series Analysis, Data Structure, Computer Science Organization, Nature Language Processing.

Columbia University New York, US
Master of Arts in Statistics *Sept 2023 – Feb. 2024*
- Overall GPA:3.71
- Main Course: Statistical Machine learning, Bayesian Statistics, Bayes Regression Multilevel.

Professional Experience

Henan Yiluo Investment CO., LTD. Shanghai, China
Analyst Intern, *Department of Strategy* Jun 2021 – Aug 2021
- Finished research reports about historical development, industry status, and business profits model of leading companies in the mechanization industry.
- Gather information about industries and their upstream and downstream, such as Baltic Freight Index for transportation industry via Wind to investigate the present developing situation and future tendency.
- Collaborated with the team to quantify the changing policy, developing circumstances as variables to adjust the business profits models and illustrated the possible influence due to the change, then to predict future revenue.
- Built a historical database of 200 stocks of mechanization industry relatively outstanding and stable in specific periods in the stock market with Python and SQL, to automatically update intraday transaction data.

Central China Securities CO., LTD. Zhengzhou, China
Asset Management Intern, Department of Strategy Dec 2020 – Jan 2021
- Worked out the industry rotation strategy by comparing fluctuation of the Hurst Exponent (a long-term memory of time series about volatility of rate of return) with the standard Bollinger Bands in a team.
- Collected the historical data from the Wind as the experiment sample, such as CSI 300, SSE Composite Index, and some industries index. And built a historical database about index values floating with time by SQL.
- Constructed a least-squares linear regression to calculate the normal range and fluctuation of Hurst Exponent with Python and found the margin of the Hurst Exponent to rotate the buy-in and sell-out strategies.
- Simulated the transaction of stocks in different industries based on the changes of Hurst Exponent, and calculated the expected rate of return, and analyzed the behavior of strategy in different industries.

Project Experience

Modeling ecDNA Copy Number Dynamics in Cancer Cell Population New York, US
Columbia Mentored Research Sept. 2024 - Now
- Simulated Moran Process to adapt to incorporate the random segregation of ecDNA during cell division.
- Employed Wasserstein metrics as distance function to conduct Approximate Bayesian Computation.
- Deducted mathematical equation of Branching Model to make it suitable for cancer cell cases.

Enhancing Intersection Safety for VRUs: Develop A Smartwatch Alert System New York, US
Columbia DitecT Lab May. 2024 – Aug. 2024
- Simulated virtual reality of NYC street view, traffic, and vehicles from a given trajectories by Unity as the training environment.
- Literature reviewed how to objectively utilize the VR to estimate effectiveness of different traffic Scenarios.
- Applied MQTT broker API in Unity and predicted the future route of a vehicle based on the current and past routes.

Legal Aid DNA Unit Research New York, US
Columbia Mentored Research Feb. 2024 – Aug. 2024
- Recognized and analyzed the statistical inference to the argument provided in STRmix Software.
- Found the inconsistency and the invalidation in potential arguments or in specific situations.
- Read the correlated published articles or reference to fully understand the concept and argument.

Data Wrangling for Relationship between News and Bitcoin New York, US
Python Project Oct 2023 – Dec 2023
- Queried the Guardian API for data with specific requirement and preprocess data to ensure its quality and integrity, such as handling the missing values, correcting inconsistencies, and standardizing data formats.
- Integrated heterogeneous data sources using SQL for seamless data flow and compatibility.
- Implemented regression model and cross validation to analyze the potential relationship between the bitcoins news headlines and the price of bitcoins.

Clustering Based on Contents and Comments of News Remote
Shanghai Jiaotong University Student Research Oct 2023 – Dec 2023

- Structuralized news articles and comments by employing doc2vec to embed texts into low-dimensional fixed-length vectors for future effectively quantifying similarity.
- Utilized single-pass clustering model to explore the implicit relation similarity between different comments which is preparation for calculating the overall similarity.

Movie Review and Sentimental Label New York, US
School Project Sept 2023 – Oct 2023

- Engineered a Bayesian-based logistic regression model by implementing a stochastic MAP estimation.
- Applied the model to IMDB movie reviews dataset, and preprocessed data by bag-of-words and TF-IDF normalization.
- Investigated the impact of batch size and learning rate on model optimization, enhancing my skills in fine-tuning machine learning models for better performance.

Prediction of Economic Situation from Economic News SA New York, US
NLP Project March 2023 – May 2023

- Scraped 200,000 news headline data and cleaned the database for removing invalid data.
- Created a Economic Relevance Filter by scikit-learn and TF-IDF with python to classify the headline.
- Applied BERT model based on the contextual pattern and locality of phrase to train the data and find their sentiment, and the model has 86% accuracy when judging the headlines' sentiment.

Missing Woman and Marriage Rate New York, US
Durf Team Research Apr 2022 – Apr 2023

- Designed a social situation based on the Ising Model about how male and female randomly interacted with each other when disproportional sex happened in rural areas and visualized the iterations to show the potential serious problems with MATLAB.
- Created simulation for the future population changes under sex discrimination situation in rural area with python.
- Assembled rural population data based on the sex, region, and time, and made a regression to find the possibly correlative variable influencing the marriage rate, such as the birth rate of boy and girl, calculated accuracy and analyzed the disturbance.

Interface Implementation Analysis New York, US
Individual Research with Java. Oct 2021 – Dec 2021

- Accomplished stack interface using linked reference structure, such as binary tree, Nodes with Java.
- Devised different sorting methods, such as Merge Sort and Quick Sort to sort the students' data such as score, rank, etc. And found the most efficient sorting method among them.

Additional Information

- **Skills:** Python(advanced), Java(intermediate), R Studio(intermediate), SQL(intermediate), MATLAB, Unity, Wind.
- **Language:** Chinese(native), English(fluent).
- **Interests:** Guitar, Swim, Chess, Hearthstone.